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Early sepsis prediction using a hybrid LSTM-GAT model: a study on the PhysioNet 2019 dataset

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ABSTRACT

Objective Sepsis is a potentially fatal systemic response to infection, in which early clinical intervention is critical to reduce mortality. This study presents a hybrid deep learning model that combines temporal and structural information from clinical data to improve early sepsis prediction.

Methods We used data from the PhysioNet/Computing in Cardiology Challenge 2019 to predict sepsis onset up to 12 hours in advance. We developed a hybrid model integrating long short-term memory (LSTM) networks and graph attention networks (GAT) to capture temporal dynamics and intervariable relationships. Performance was compared with three baseline models. To ensure robustness, all models were trained using five repeated train-test splits with different random seeds.

Results The dataset includes 40 336 adult ICU patients. Of all the patients, 2932 developed sepsis during their stay. Each patient's data includes hourly data on 40 clinical variables, including vital signs, laboratory results and demographic information. The LSTM-GAT model achieved an area under the receiver operating characteristic curve (AUROC) of 0.853±0.005, an F1-score of 0.627±0.006 and a specificity of 0.872±0.007, outperforming baseline models. Despite being trained on fixed temporal windows, the model generalised well across multiple prediction horizons without retraining.

Discussion By integrating temporal and structural representations, the proposed approach achieves improved predictive performance compared with the baseline. This capability may support earlier identification of high-risk patients and enhance timely clinical decision-making in critical care environments.

Conclusions The proposed model demonstrates the advantage of combining sequence and graph-based methods. It offers a promising tool for real-time clinical decision support in sepsis detection.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Sepsis is a major global health crisis. Early diagnosis and intervention are critical to reduce mortality.
⇒ Machine learning algorithms are widely used for sepsis prediction but often rely on hand-crafted features (eg, average heart rate over 6 hours).

WHAT THIS STUDY ADDS

⇒ This paper introduces a hybrid deep learning model (long short-term memory (LSTM) networks and graph attention networks (GAT)) that learns from raw time-series data.
⇒ The model outperforms standard LSTM and GAT baselines in predicting sepsis up to 12 hours before onset.
⇒ The proposed method generalises across different prediction windows without retraining.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ The proposed LSTM-GAT model offers a clinically deployable approach for real-time sepsis detection in ICU settings, operating efficiently without GPU requirements.
⇒ Its ability to generalise across multiple prediction horizons without retraining may reduce model maintenance burden and support practical implementation in clinical environments.

INTRODUCTION

Sepsis is a severe inflammatory response to infection that can lead to organ failure and death if not detected early. It causes one in five deaths globally and affects up to 50 million people annually, with the highest incidence and mortality in low- and middle-income countries.[1–3] It places a substantial burden on healthcare systems, often necessitating intensive care unit (ICU) admission and costly interventions. The WHO estimates that 15 in 1000 hospitalised patients develop sepsis,[1] consuming up to 2.65% of healthcare budgets.[4] Patients with sepsis also experience longer ICU stays and higher rehospitalisation rates.[5]

Early detection is challenging due to its ambiguous symptoms and patient variability.[6] Prompt diagnosis and intervention are critical, as each hour of treatment delay increases mortality risk.[7 8] Existing sepsis scoring systems, such as the sequential organ failure assessment (SOFA) and systemic inflammatory response syndrome (SIRS), have limited predictive performance, partly due to their reliance on static thresholds.[6 9] Unlike conventional static scoring systems, machine learning (ML) models enable continuous

analysis of patient data, supporting dynamic risk assessment and earlier detection of clinical decline.[10 11]

This increasing interest in ML-based approaches has been highlighted by benchmark efforts like the PhysioNet/Computing in Cardiology 2019 Challenge.[12] Notably, top-performing models used gradient boosted decision trees, specifically XGBoost.[13–15] Despite strong performance, their reliance on extensive feature engineering and high computational complexity limits their real-world applications.

Bloch *et al*[16] applied a support vector machine (SVM) using changes in four vital signs, achieving an area under the receiver operating characteristic curve (AUROC) of 0.88. Nemati *et al*[17] proposed the Artificial Intelligence Sepsis Expert algorithm, combining standard electronic medical record data with high-frequency physiological signals, reaching an area under the curve (AUC) of 0.83–0.85. Recently, Liu *et al*[18] evaluated eight ML models across two feature sets, finding that adding demographic and medical history improved performance, with gradient boosting achieving the highest AUC.

Deep learning models capture temporal dependencies and nonlinear patterns without manual feature engineering. Hybrid models integrating convolutional neural networks (CNNs) with long short-term memory (LSTM) architectures have reported AUROCs exceeding 0.85 on electronic health records data.[19 20]

Recurrent neural networks (RNNs) and variants like gated recurrent units (GRUs) have been widely used to model temporal dynamics in sepsis prediction, typically achieving AUROCs around 0.80; however, limitations such as low specificity, suboptimal F1-scores and limited external validation persist.[21–23] Recent work explored Gaussian process-based imputation with RNNs,[24] as well as transformer-based models with reported AUROCs of 0.846 for internal validation and 0.807 for external validation.[25] Notably, COMPOSER, a clinically evaluated model, demonstrated reduced mortality and improved sepsis bundle compliance.[26 27]

Graph-based models can capture structural relationships between clinical variables. Lee *et al*[28] represented clinical variables as nodes, and edges were defined by statistical correlations. While effective in capturing feature dependencies, this model lacked temporal modelling. Conversely, CNN-LSTM models handle temporal trends but often overlook interfeature interactions.

To overcome these limitations, we developed a hybrid deep learning approach that integrates LSTM networks and graph attention networks (GAT). The model provides early sepsis warnings to ICU clinicians. It jointly captures temporal dynamics and intervariable dependencies in clinical time-series data. In contrast to earlier approaches that relied heavily on manually engineered features or focused on a single aspect of the data structure, our method enables more comprehensive representation learning. We developed and evaluated several model variants to assess the contribution of each component.

METHODS

Data

We used the PhysioNet/Computing in Cardiology Challenge 2019 dataset,[12] including ICU data from 40 336 adult patients in two US hospitals. Each patient has 40 clinical features recorded hourly, including 8 vital signs, 6 demographic variables and 26 lab tests. Missing values, due to irregular testing, are presented as 'NaN'. Participant characteristics by sepsis status are shown in table 1.

Following Sepsis-3 guidelines, the onset of sepsis was characterised by the co-occurrence of suspected infection and a two-point increase in the SOFA score. The dataset provides an hourly sepsis label. For patients developing sepsis, the time of sepsis occurrence is also provided, and the label is set to one starting from 6 hours before documented onset.

In the original PhysioNet Challenge,[12] predictions up to 12 hours before onset were rewarded, with the highest reward at 6 hours before. We extended the positive label window to 12 hours before the onset of sepsis. Therefore, results are not directly comparable to studies using the default labelling strategy.

Data preprocessing

We applied several preprocessing steps to ensure data consistency across all participants. Among the 40 clinical variables in the PhysioNet dataset,[12] 13 features (32.5%) were excluded due to extremely high missingness (approximately 85%–95%) and limited measurement

Table 1 Baseline characteristics of ICU patients stratified by sepsis status

Variable	Sepsis (n=2932)	No sepsis (n=37404)
Age (mean±SD)	62.06±16.38	61.61±16.49
Gender	Male: 59.3%	Male: 55.7%
	Female: 40.7%	Female: 44.3%
ICU length of stay (mean±SD)	59.42±59.45	37.41±15.96
Medical ICU admission	55.80%	48.70%
Surgical ICU admission	44.20%	51.30%

Continuous variables are presented as mean (SD), and categorical variables as percentages.
ICU, intensive care unit.

frequency within ICU trajectories. However, some clinically relevant variables were retained despite higher missingness since they demonstrated importance in preliminary analyses. The shock index was introduced as a derived feature, calculated as the ratio of heart rate to systolic blood pressure. This resulted in 28 features for model training, comprising 7 vital signs, 14 laboratory variables, 6 demographic and ICU-related variables and 1 derived feature (online supplemental Table 1).

Missing values were addressed using a two-step procedure. Initially, forward-fill imputation was performed within each patient's time series. Remaining missing values were handled using feature-specific imputation; categorical variables were imputed using the mode when no earlier observation was present and numerical variables using multivariate imputation by chained equations. All numerical features were standardised using training-set statistics to prevent data leakage.

Sequential data were prepared using a sliding window approach. For patients with sepsis, overlapping windows of fixed length (10 hours) were employed, capturing preonset dynamics. In contrast, for patients without sepsis, non-overlapping windows were used to avoid redundancy. Sequences shorter than the window size were zero-padded at the beginning to preserve the most recent observations. This approach enabled fixed-length input windows for all patients and outperformed repeating the last observation.

Each time window was labelled, indicating whether sepsis had occurred or would occur within the following 12 hours. In addition to the label, the exact timing of sepsis onset (if present) and the time of the last observation within the window were stored for utility function calculation. Class imbalance was addressed by random downsampling of the majority (non-sepsis) class. Synthetic oversampling (eg, Synthetic Minority Oversampling Technique (SMOTE)) was not applied in this study because synthetic interpolation between time-series windows may distort physiological trajectories.

Model architectures

We developed several deep learning models to predict sepsis onset using 10-hour clinical windows, comprising 28 features per hour. The implemented models included LSTM, GAT, a hybrid LSTM-GAT model and a temporal variant referred to as temporal GAT. The GAT-based models incorporated a static, fully connected graph structure in which each node represents a feature and edges are established between all pairs of nodes.

LSTM networks

LSTM networks[29] are a type of RNN designed to capture long-range dependencies in sequential data. In this study, LSTMs serve as a baseline to model the temporal evolution of clinical features over 10-hour windows for early sepsis prediction.

GAT architecture

To model interfeature dependencies, the GAT architecture[30] was used, where each node represents a clinical feature. The model receives the temporal trajectory of each feature over a 10-hour window, and attention weights are applied over the fully connected static feature graph, allowing aggregation of relevant information and learning of structural relationships.

LSTM-GAT (temporal-spatial hybrid)

The LSTM-GAT model first applies LSTM layers to model the temporal change of the clinical features over a 10-hour window. These representations are then fed into a fully connected graph, where a GAT captures spatial dependencies among features, enabling learning of both temporal dynamics and interfeature relationships.

Temporal GAT

Unlike the previous architectures, the temporal GAT models time steps as graph nodes and uses directed edges to learn the changes in clinical variables over time, evaluating whether attention mechanisms improve temporal modelling.

Model training and evaluation

The dataset was randomly split into training (70%), validation (15%) and test (15%) sets. Models were trained using the Adam optimiser (batch size=32 and learning rate=0.001) for up to 100 epochs, with early stopping on validation loss. Hyperparameters were optimised using the validation set. Class imbalance was addressed through random downsampling of the majority class. Each experiment was repeated with five different random seeds, and the average metrics are reported.

Model performance was evaluated using AUROC, area under the precision-recall curve (AUPRC) and the PhysioNet utility score,[12] which rewards earlier detection and assigns the highest score to predictions made 6 hours before sepsis onset. We also report the F1-score, precision, recall and specificity. Thresholds were selected using Youden's J statistic to maximise the sum of sensitivity and specificity.

Further analytical analyses

To better understand the model's mechanisms and temporal performance, we conducted additional analyses: (1) evaluation of model performance using vital signs only, (2) applying SHAPGradient to interpret feature contributions, (3) temporal analysis of prediction outcomes across multiple intervals within the 12 hours before sepsis onset and (4) sensitivity-specificity threshold analysis to examine clinically relevant operating points.

Vital sign-only model evaluation

The dataset includes 8 vital sign variables, 26 laboratory features and 6 demographic variables. However, the laboratory features have high percentages of missing values, and the demographic variables are largely static. To evaluate the predictive capacity of vital signs, we retrained the best-performing model (LSTM-GAT) using only these features, providing a baseline for comparison with the full-feature model.

Feature attribution

Feature contributions were analysed using SHAPGradient, a gradient-based extension of SHapley Additive exPlanations (SHAP) designed specifically for neural networks. Compared with traditional SHAP methods, which can be computationally intensive or inefficient for gradient-based models, SHAPGradient leverages backpropagation to efficiently estimate feature importance.

Temporal analysis of prediction outcomes

To investigate the temporal dynamics of the model's predictive performance, we analysed its outputs within the 12-hour window preceding sepsis onset, divided into six separate 2-hour intervals (ie, 10–12 hours, 8–10 hours,….,0–2 hours before onset). For each interval, true positive and false negative rates were calculated to assess how the model's sensitivity varied across time and identify the interval in which it most effectively detected sepsis.

Sensitivity-specificity threshold analysis

Threshold analyses were performed on four laboratory variables (pH, creatinine, white blood cell (WBC) and platelets) using 200 candidate values across a physiologically relevant range. Predictions were generated using univariate thresholds, and sensitivity and specificity were computed at each point. Optimal cutoffs were defined using Youden's Index, and performance was also examined at a predefined 90% sensitivity operating point to represent a high-sensitivity early-warning scenario.

External validation

To assess robustness across institutions, external validation was performed using the Medical Information Mart for Intensive Care III (MIMIC-III).[31] Because laboratory measurements were largely missing, analyses were restricted to vital signs. Variables were aligned with PhysioNet[12] feature definitions and standardised using training-set parameters. Sepsis onset was reconstructed using Sepsis-3-consistent criteria, combining suspected infection (antibiotics plus blood cultures) with a ≥2-point rise in a partial SOFA score. Validation thresholds were fixed based on the PhysioNet dataset.[12]

Data and code availability and conflicts of interest

The data used in this study include the publicly available PhysioNet/Computing in Cardiology Challenge 2019 dataset[12] (https://physionet.org/content/challenge-2019/1.0.0/) and the MIMIC-III clinical[31] (V.1.4; https://physionet.org/content/mimiciii/1.4/). The code used for analysis and model development is available at: https://github.com/bk-ai-health/sepsis-lstm-gat.

Outcome

This study used the publicly available, de-identified datasets, including the PhysioNet 2019 Challenge dataset[12] and the MIMIC-III clinical database (V.1.4).[31] Access to MIMIC-III[31] requires completion of the PhysioNet credentialing process and a data use agreement. For questions regarding the datasets, contact PhysioNet at physionet@mit.edu.

RESULT

This section presents the performance of LSTM, feature-level GAT, an end-to-end LSTM–GAT hybrid, and temporal GAT models for early sepsis prediction. All models were trained and evaluated five times with different random seeds, and average metrics with SD are reported in table 2. Figure 1 shows ROC and precision-recall curves for comparison. In addition to standard metrics, we report the false-alarm rate (1–specificity) to reflect potential alert burden in clinical deployment.

Feature importance analysis using SHAPGradient revealed the top 10 predictors for early sepsis detection. As illustrated in figure 2, Unit2, Unit1 and temperature were the most influential features, highlighting the importance of both physiological variables and clinical context (eg, ICU stay length) in predicting sepsis. Given that 6 of the 10 top features were vital signs, we evaluated the model's predictive power using only these inputs, excluding laboratory and demographic variables. The aim was to assess the predictive value of vital signs and provide a reference point for evaluating the contribution of additional feature types. As expected, performance declined compared with the full feature set (online supplemental Table 2), with AUROC dropping from 0.853 to 0.730 and the utility score decreasing from 0.705 to 0.584.

Table 2 Comparative predictive performance of the evaluated models for early sepsis detection, reported as mean (SD) across five runs with different random seeds

Model	F1-score	Precision	Recall (sensitivity)	Specificity	False alarm rate (1–specificity)	Utility score
LSTM	0.608±0.009	**0.568±0.006**	0.654±0.013	0.782±0.072	0.218±0.072	0.688±0.012
GAT	0.509±0.002	0.441±0.012	0.602±0.022	0.817±0.016	0.183±0.016	0.630±0.016
LSTM-GAT	**0.627±0.006**	0.565±0.012	**0.706±0.009**	**0.872±0.007**	**0.128±0.007**	**0.705±0.010**
Temporal GAT	0.410±0.015	0.313±0.032	0.603±0.059	0.677±0.081	0.323±0.081	0.619±0.039

Bold values indicate the best-performing model for each metric.
GAT, graph attention networks; LSTM, long short-term memory.

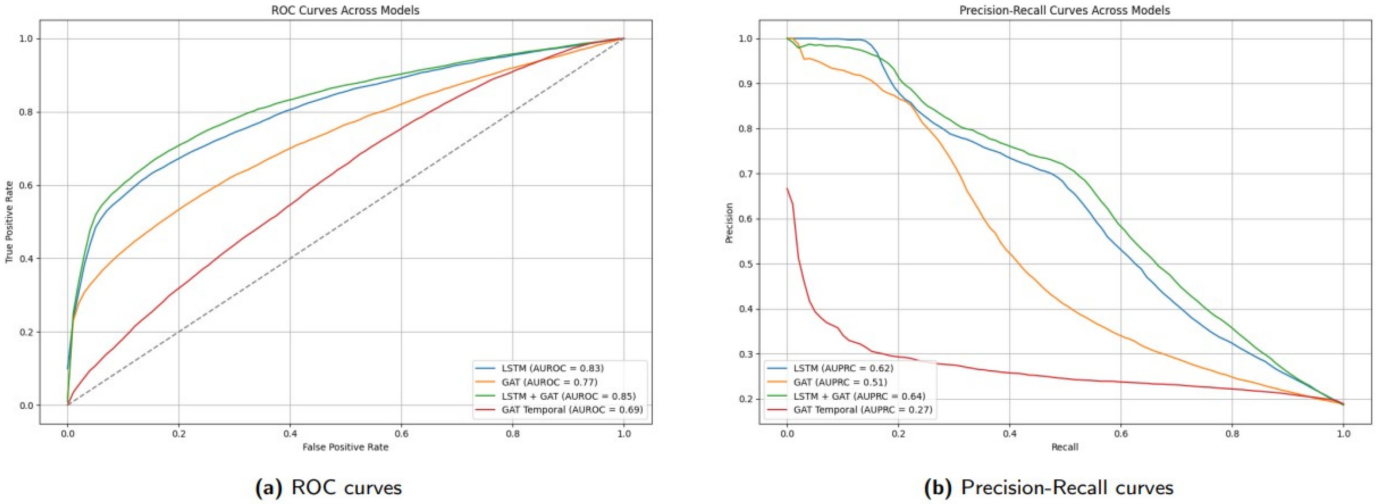


Figure 1 (a) ROC curves and (b) precision–recall curves comparing the predictive performance of the LSTM, GAT, LSTM-GAT and temporal GAT models for early sepsis detection. The LSTM-GAT model demonstrates superior discrimination and precision–recall balance relative to the baseline architectures. GAT, graph attention networks; LSTM, long short-term memory; ROC, receiver operating characteristic.

To examine the model's temporal performance as sepsis onset approached, we analysed prediction outcomes across six consecutive 2-hour intervals within the 12-hour period preceding onset. As illustrated in online supplemental figure 1 and figure 3, sensitivity did not consistently improve as the interval became closer to sepsis onset. Interestingly, the highest sensitivity occurred between 6 and 8 hours before onset, with a slight decrease in the final 0–2 hour interval. This pattern reveals that performance does not necessarily improve closer to clinical onset.

Sensitivity and specificity analyses were conducted to derive clinically interpretable thresholds for selected laboratory variables (online supplemental Table 3). Based

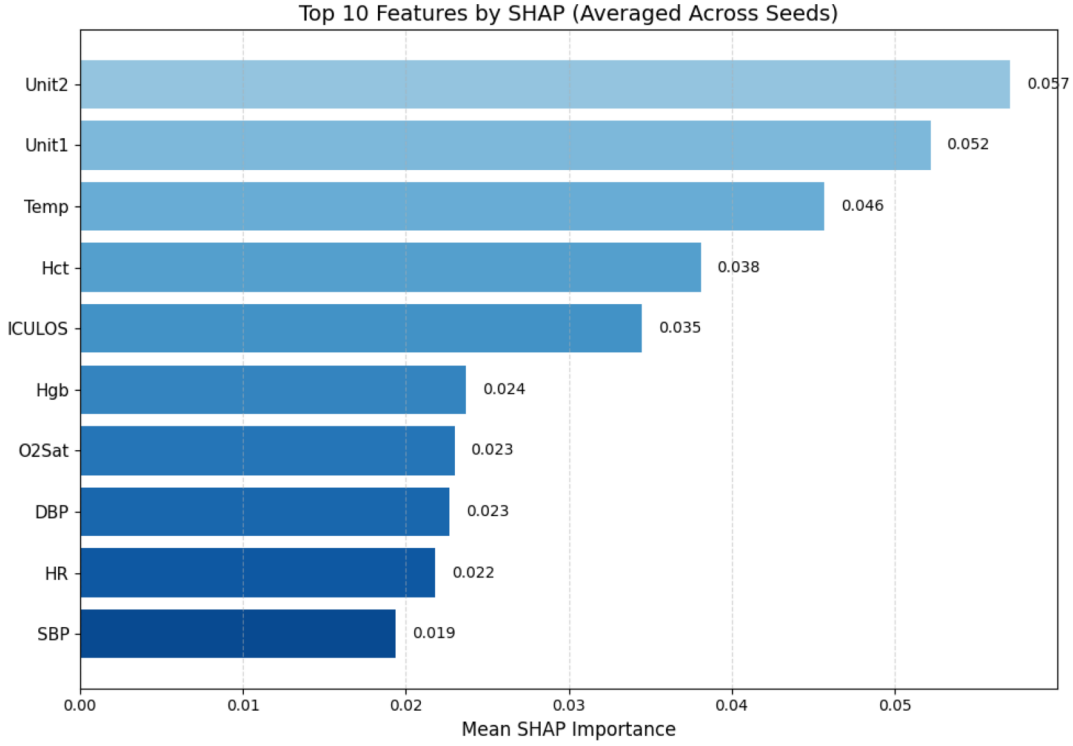


Figure 2 Top 10 features contributing to early sepsis prediction identified using SHAPGradient, averaged across five runs with different random seeds. Higher SHAP values indicate greater influence on model predictions, highlighting the importance of both physiological measurements and ICU-related variables. DBP, diastolic blood pressure; Hct, haematocrit; Hgb, haemoglobin; HR, heart rate; ICU, intensive care unit; ICULOS, ICU length of stay; O_2Sat, oxygen saturation; SBP, systolic blood pressure; SHAP, SHapley Additive exPlanations.

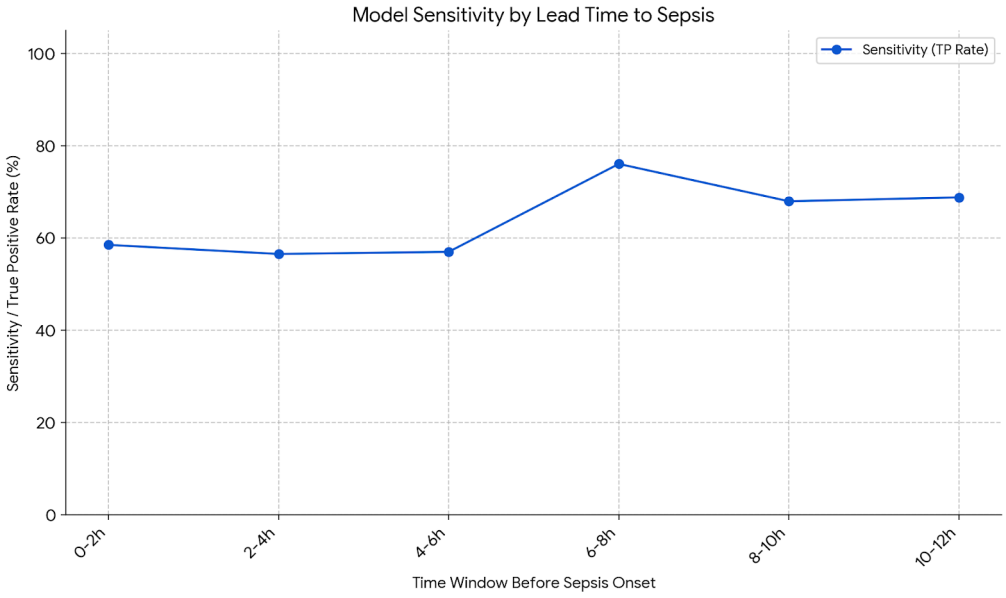


Figure 3 Sensitivity (true positive rate) of the LSTM-GATmodel across consecutive 2-hour intervals within the 12 hours preceding sepsis onset, illustrating the model's ability to detect high-risk patients at varying temporal distances from clinical onset. LSTM, long short-term memory; GAT, graph attention networks.

on Youden-index optimisation, pH demonstrated higher specificity, while creatinine achieved the highest sensitivity among the evaluated laboratory thresholds (0.707). Platelet count showed an intermediate trade-off, and WBC exhibited a more balanced performance profile. Lactate was excluded from threshold optimisation due to unstable operating points but remained part of the predictive modelling. At a predefined 90% sensitivity operating point, specificity decreased to 0.439 (false-positive rate 56%), consistent with the expected trade-off.

Since laboratory measurements were highly missing in MIMIC-III,[31] external validation was restricted to vital signs. Compared with the internal vital-sign-only AUROC of 0.730, the external AUROC was 0.70, with an AUPRC of 0.25. Recall remained relatively high (0.73), whereas precision declined significantly, reflecting differences in sepsis prevalence, label definitions and measurement practices between MIMIC-III[31] and PhysioNet[12] cohorts. These findings are consistent with variations in clinical populations and measurement settings.

DISCUSSION

We proposed and evaluated deep learning models for early sepsis prediction, focusing on temporal dynamics and feature interdependencies. The LSTM-GAT hybrid achieved the best overall performance (AUROC, F1-score, recall, specificity, utility score and AUPRC). Its high AUPRC reflects a strong precision-recall balance despite class imbalance. Consistent with these findings, the hybrid LSTM-GAT model also achieved a lower false-alarm rate compared with the baseline models, suggesting improved clinical usability by potentially reducing unnecessary alerts while maintaining strong sensitivity. In addition, the mean inference time per 10-hour window was 2.29±0.29 ms (median 2.12 ms) on a CPU (Intel Core i9-11900, no GPU required), corresponding to more than 400 windows processed per second, indicating suitability for real-time clinical decision support.

The standard LSTM model showed competitive performance, particularly excelling in precision and AUPRC, but had lower recall and specificity, indicating more missed and false-positive cases. This suggests the added benefit of incorporating graph-based attention. GAT and temporal GAT models underperformed. The poor performance of the temporal GAT suggests that modelling time as graph nodes is less effective than capturing feature-level temporal patterns.

To evaluate our findings, we compared our results with previous studies targeting early sepsis prediction across multiple datasets and architectures; a broader multimetric comparison is summarised in online supplemental Table 4.[18–20 22 32 33] Reported AUROCs include 0.83 by Kim *et al*,[32] 0.79 by Scherpf *et al*[20] and 0.83 by Wickramaratne *et al*.[22] Our LSTM-GAT model outperformed all, achieving an AUROC of 0.853, with strong recall (0.706) and specificity (0.872). Kim *et al* reported higher sensitivity, but their model used data from a single hospital with lower missingness, whereas our model was trained from two hospitals with substantial missingness, making the task more challenging.[21]

Feature importance analysis showed that six of the top 10 important predictors were vital signs (temperature (Temp), oxygen saturation (O_2Sat), diastolic blood pressure (DBP), heart rate (HR) and systolic blood pressure (SBP)), along with two lab values (haematocrit (Hct) and haemoglobin (Hgb)) and three contextual variables:

Unit1 (medical ICU), Unit2 (surgical ICU) and ICU length of stay (ICULOS). These ICU-specific variables likely reflect baseline risk differences, with ICULOS indicating higher sepsis risk from prolonged ICU stays.

These findings align with prior studies where SBP, HR, Temp and respiratory rate (RR) were frequently identified as important. Hct and blood urea nitrogen also appeared consistently.[22 32] Clinically, these features align with major domains of sepsis pathophysiology. Temperature dysregulation is consistent with established sepsis criteria, while changes in HR, SBP and DBP may reflect haemodynamic instability. Reduced O_2Sat suggests early respiratory dysfunction, and variations in Hct and Hgb may reflect haemodilution or physiological stress observed in septic patients. In contrast, variables like creatinine, WBC and lactate were not top-ranked here, possibly due to dataset or preprocessing differences. Our model prioritised less commonly reported features such as pH and O_2Sat, suggesting the model's ability to capture subtle physiological patterns.

To examine the contribution of vital signs, we compared the LSTM-GAT model's performance using all features versus only vital signs. The vital sign-only model showed a notable drop in AUROC, F1-score and utility score, indicating that while vital signs are crucial components, laboratory and demographic features significantly enhance predictive power. Rangan *et al*[34] found that combinations like (HR+Temp) and (RR+Temp) yielded the highest AUROCs among vital sign pairs. Liu *et al*[17] also showed that adding demographic and clinical history features to vital signs improved overall model performance.

In contrast to earlier research that retrained models and redefined sepsis labels for different time intervals,[20 32] we trained a single model to predict sepsis 12 hours ahead and retrospectively evaluated its performance across time windows. Kim *et al*[32] reported improved AUROC values as prediction time approached, and Scherpf *et al*[20] observed an increase from 0.76 at 12 hours to 0.81 at 3 hours. However, our approach reflects a more realistic clinical scenario in which the model must generalise across time without retraining.

The highest true positive rate in our model occurred 6–8 hours before onset, followed by a slight decline. This suggests the model captured the strongest predictive signals early in the sepsis trajectory. Besides, features significant in sepsis prediction may differ across time windows. The model's hyperparameters were not validated for each 2-hour interval, possibly optimally tuned for a specific window but less so for others.

External validation on the independent MIMIC-III cohort[31] demonstrated a lower AUROC, likely reflecting differences in sepsis prevalence, documentation practices and data acquisition protocols between datasets. These findings highlight the importance of site-specific threshold calibration before clinical deployment.

This study was conducted retrospectively using data from two hospitals and the MIMIC-III clinical database,[31] which may limit immediate generalisability to other healthcare systems. Sepsis labels were defined using Sepsis-3 criteria and may be affected by documentation timing and measurement frequency. Despite stable performance across multiple prediction horizons, hyperparameters were not optimised separately for each interval. In addition, feature attributions obtained from SHAPGradient reflect model behaviour rather than causality and should be interpreted with caution.

CONCLUSION

In this study, we proposed a hybrid LSTM-GAT model for early sepsis prediction and evaluated its performance on the PhysioNet 2019 Challenge dataset.[12] The results showed strong and balanced predictive performance, outperforming baseline architectures. Our model also demonstrated robust performance across different time intervals preceding sepsis onset without retraining, which supports its potential for real-world clinical application. Feature importance analysis revealed clinically meaningful variables, including vital signs, laboratory values and ICU-related demographic factors. These findings highlight the importance of integrating temporal and sequential patterns in clinical data for accurate early sepsis detection.

Contributors BK conceived and designed the study, performed data preprocessing, developed and trained the models, conducted the statistical analyses and drafted the manuscript. SK contributed to study design, provided methodological guidance and critically revised the manuscript for important intellectual content. Both authors reviewed and approved the final version of the manuscript. BK is the guarantor of this work and accepts full responsibility for the integrity of the study, the accuracy of the data analysis and the decision to submit for publication. We used OpenAI's GPT-4 (via ChatGPT) for editorial support. It was mainly used for improving clarity and grammar. I have also used it for increasing the readability of sentences or suggesting keywords.

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Competing interests None declared.

Patient consent for publication Not applicable.

Ethics approval Informed consent and ethical approval were obtained by the original data providers, and no new participant data were collected by the authors.

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Data availability statement Data are available in a public open access repository. The data used in this study include the publicly available PhysioNet/Computing in Cardiology Challenge 2019 dataset (https://physionet.org/content/challenge-2019/1.0.0/) and the MIMIC-III clinical (V.1.4; https://physionet.org/content/mimiciii/1.4/). The code used for analysis and model development is available at: https://github.com/bk-ai-health/sepsis-lstm-gat

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REFERENCES

1 World Health Organization. Sepsis. 2020. Available: https://www.who.int/news-room/fact-sheets/detail/sepsis
2 Rudd KE, Johnson SC, Agesa KM, *et al*. Global, regional, and national sepsis incidence and mortality, 1990–2017: analysis for the Global Burden of Disease Study. *The Lancet* 2020;395:200–11.
3 Bauer M, Gerlach H, Vogelmann T, *et al*. Mortality in sepsis and septic shock in Europe, North America and Australia between 2009 and 2019- results from a systematic review and meta-analysis. *Crit Care* 2020;24:239.
4 van den Berg M, van Beuningen FE, Ter Maaten JC, *et al*. Hospital-related costs of sepsis around the world: A systematic review exploring the economic burden of sepsis. *J Crit Care* 2022;71:154096.
5 Farrah K, McIntyre L, Doig CJ, *et al*. Sepsis-Associated Mortality, Resource Use, and Healthcare Costs: A Propensity-Matched Cohort Study. *Crit Care Med* 2021;49:215–27.
6 Haas R, McGill SC. Artificial Intelligence for the Prediction of Sepsis in Adults. *cjht* 2022;2.
7 Goh KH, Wang L, Yeow AYK, *et al*. Artificial intelligence in sepsis early prediction and diagnosis using unstructured data in healthcare. *Nat Commun* 2021;12:711.
8 Bomrah S, Uddin M, Upadhyay U, *et al*. A scoping review of machine learning for sepsis prediction- feature engineering strategies and model performance: a step towards explainability. *Crit Care* 2024;28:180.
9 Yadgarov MY, Landoni G, Berikashvili LB, *et al*. Early detection of sepsis using machine learning algorithms: a systematic review and network meta-analysis. *Front Med (Lausanne)* 2024;11:1491358.
10 Islam MM, Nasrin T, Walther BA, *et al*. Prediction of sepsis patients using machine learning approach: A meta-analysis. *Comput Methods Programs Biomed* 2019;170:1–9.
11 Fleuren LM, Klausch TLT, Zwager CL, *et al*. Machine learning for the prediction of sepsis: a systematic review and meta-analysis of diagnostic test accuracy. *Intensive Care Med* 2020;46:383–400.
12 Reyna M, Josef C, Jeter R, *et al*. Early prediction of sepsis from clinical data: the physionet/computing in cardiology challenge 2019 (version 1.0.0). *PhysioNet* 2019.
13 Morrill J, Kormilitzin A, Nevado-Holgado A, *et al*. The signature-based model for early detection of sepsis from electronic health records in the intensive care unit. 2019 Computing in Cardiology Conference; IEEE, 1.
14 Du JA, Sadr N, Chazal P. Automated prediction of sepsis onset using gradient boosted decision trees. 2019 Computing in Cardiology (CinC); 2019:1.
15 Zabihi M, Kiranyaz S, Gabbouj M. Sepsis prediction in intensive care unit using ensemble of xgboost models. 2019 Computing in Cardiology Conference; 2019:1.
16 Bloch E, Rotem T, Cohen J, *et al*. Machine Learning Models for Analysis of Vital Signs Dynamics: A Case for Sepsis Onset Prediction. *J Healthc Eng* 2019;2019:5930379.
17 Nemati S, Holder A, Razmi F, *et al*. An Interpretable Machine Learning Model for Accurate Prediction of Sepsis in the ICU. *Crit Care Med* 2018;46:547–53.
18 Liu Z, Shu W, Li T, *et al*. Interpretable machine learning for predicting sepsis risk in emergency triage patients. *Sci Rep* 2025;15:887.
19 Lauritsen SM, Kalør ME, Kongsgaard EL, *et al*. Early detection of sepsis utilizing deep learning on electronic health record event sequences. *Artif Intell Med* 2020;104:101820.
20 Rafiei A, Rezaee A, Hajati F, *et al*. SSP: Early prediction of sepsis using fully connected LSTM-CNN model. *Comput Biol Med* 2021;128:104110.
21 Scherpf M, Gräßer F, Malberg H, *et al*. Predicting sepsis with a recurrent neural network using the MIMIC III database. *Comput Biol Med* 2019;113:103395.
22 Kim T, Tae Y, Yeo HJ, *et al*. Development and Validation of Deep-Learning-Based Sepsis and Septic Shock Early Prediction System (DeepSEPS) Using Real-World ICU Data. *JCM* 2023;12:7156.
23 Wickramaratne SD, Shaad Mahmud M. Bi-directional gated recurrent unit based ensemble model for the early detection of sepsis. 2020 42nd Annual International Conference of the IEEE Engineering in Medicine &Amp; Biology Society (EMBC); IEEE, 70–3. Montreal, QC, Canada.
24 Futoma J, Hariharan S, Heller K, *et al*. An improved multi-output gaussian process rnn with real-time validation for early sepsis detection. Machine Learning for Healthcare Conference; 2017:243–54.
25 Moor M, Bennett N, Plečko D, *et al*. Predicting sepsis using deep learning across international sites: a retrospective development and validation study. *EClinicalMedicine* 2023;62:102124.
26 Shashikumar SP, Josef C, Sharma A, *et al*. DeepAISE--An end-to-end development and deployment of a recurrent neural survival model for early prediction of sepsis. *arXiv* 2019.
27 Boussina A, Shashikumar SP, Malhotra A, *et al*. Impact of a deep learning sepsis prediction model on quality of care and survival. *NPJ Digit Med* 2024;7:14.
28 Lee BT, Kwon OY, Park H, *et al*. Graph Convolutional Networks-Based Noisy Data Imputation in Electronic Health Record. *Crit Care Med* 2020;48:e1106–11.
29 Hochreiter S, Schmidhuber J. Long short-term memory. *Neural Comput* 1997;9:1735–80.
30 Veličković P, Cucurull G, Casanova A, *et al*. Graph attention networks. *arXiv* 2017.
31 Johnson A, Pollard T, Mark R. Data from: MIMIC-III Clinical Database (version 1.4). *PhysioNet* 2016.
32 Kim JK, Ahn W, Park S, *et al*. Early Prediction of Sepsis Onset Using Neural Architecture Search Based on Genetic Algorithms. *Int J Environ Res Public Health* 2022;19:2349.
33 Yang M, Liu C, Wang X, *et al*. An Explainable Artificial Intelligence Predictor for Early Detection of Sepsis. *Crit Care Med* 2020;48:e1091–6.
34 Rangan ES, Pathinarupothi RK, Anand KJS, *et al*. Performance effectiveness of vital parameter combinations for early warning of sepsis-an exhaustive study using machine learning. *JAMIA Open* 2022;5.